



SplatSuRe: Selective Super-Resolution for Multi-view Consistent 3D Gaussian Splatting

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Motivation

Can we use 2D super-resolution to build high-resolution 3DGS models but also remain consistent to the ground truth?

Low Resolution

SRGS



21.69 PSNR 0.266 LPIPS



23.11 PSNR 0.234 LPIPS

Ours

Ground Truth

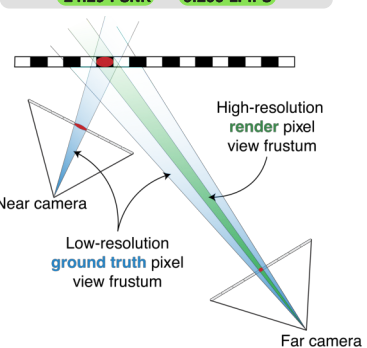


24.29 PSNR 0.206 LPIPS



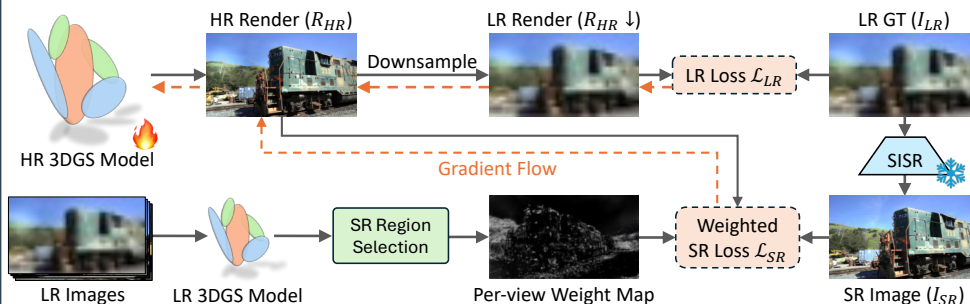
Images of a scene do not sample 3D content uniformly

Near cameras can provide high-resolution information for rendering distant views



We can use this disparity in captured frequency across views to inform where super-resolution is really required

Method



Gaussian Fidelity Score

ρ = Ratio of Gaussian's max to min scene-space radius across views

$\rho \approx$ Disparity in sampling across views

High $\rho \rightarrow$ High variance in sampled frequency

Low $\rho \rightarrow$ Uniform sampling

score = Sigmoid($\frac{\rho - \tau}{k}$), where τ is a threshold and k controls smoothness

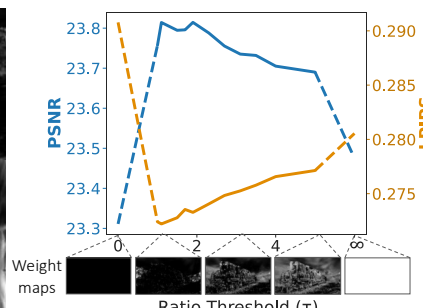
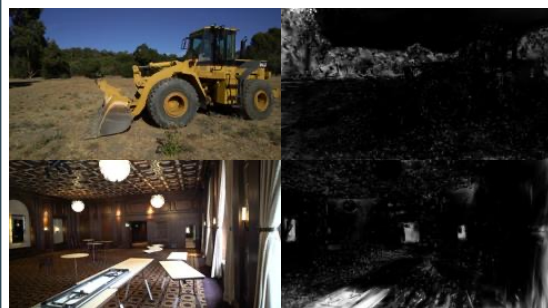
Super-Resolution Region Selection

Convert per-Gaussian score to per-view weight map for super-resolution selection

$$W_t = \left(1 - \text{Render}(\text{score}_G)\right) + \text{Render}(\mathbf{1}_{M(t)}(\mathcal{G}))$$

Regions with low fidelity scores Regions closest to current view

$\mathcal{L}_{SR}$ is spatially weighted by this per-view weight map W_t



Results

Mip-Splatting

SRGS

Ours

GT



We achieve consistently sharper novel renders and improved consistency with the ground truth by selectively applying super-resolution!

Method	Tanks & Temples (rendered at 4x input resolution)							
	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	CMMD $\downarrow$	DreamSim $\downarrow$	MUSIQ $\uparrow$	NIQE $\downarrow$
3DGS (LR)	0.669	19.41	0.350	71.58	2.013	0.0895	57.776	3.412
3DGS + StableSR	0.751	22.47	0.300	59.29	1.123	0.0667	56.748	4.945
Mip-Splatting	0.767	23.10	0.303	52.46	1.137	0.0597	46.571	5.043
SRGS + StableSR	0.771	23.32	0.286	49.11	1.048	0.0535	55.209	4.633
Ours + StableSR	0.784	23.81	0.272	37.72	1.040	0.0413	58.332	3.928

* Please see our website and paper for additional qualitative and quantitative results

Acknowledgements

This work was made possible by NSF Grants 21-37229 and 22-35050, DARPA T1AMAT, and the NSF TRAILS Institute (2229885). This research is based upon work supported by the Office of the Director of National Intelligence (ODNI), Intelligence Advanced Research Projects Activity (IARPA), via IARPA R&D Contract No. 140D0423C0076. The views and conclusions contained herein are those of the authors and should not be interpreted as necessarily representing the official policies or endorsements, either expressed or implied, of the ODNI, IARPA, or the U.S. Government. The U.S. Government is authorized to reproduce and distribute reprints for Governmental purposes notwithstanding any copyright annotation thereon. Additional support was provided by Coefficient Giving.